PMHIF-Net: A Prior-Guided Mamba Hierarchical Interactive Fusion Network for Hyperspectral and Multispectral Image Fusion

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***Abstract*—Hyperspectral and multispectral image fusion (HMIF) aims to generate hyperspectral images (HSIs) with high-spatial resolution and favorable spectral fidelity. Benefiting from efficient and powerful global modeling capacity, Mamba has become a prevailing solution for the HMIF task in recent years. Nevertheless, most existing Mamba-based fusion methods fail to effectively capture inherent characteristics, including spatial edge structures and spectral reflectance variation rates during state-space modeling, and suffer from redundant information in the fusion pipeline. To address these limitations, this article proposes a prior-guided hierarchical interactive Mamba fusion network termed PMHIF-Net. First, edge priors and spectral gradient priors are embedded into the state-space model (SSM), and the spatial learning module (SpaLM) and spectral learning module (SpeLM) are developed. The priors explicitly participate in state-space evolution to capture local spatial edge details and spectral variation patterns across adjacent bands. Second, to correlate prior-enhanced features and enable progressive optimization of multilevel features, the spatial–spectral interactive Mamba block (SSIMB) is constructed. Deep cross-modal interaction is realized via bidirectional dynamic interaction and mutual control mechanisms. Finally, the refined gated reconstruction module (RGRM) is designed to couple and refine hierarchical features. Frequency-domain priors are introduced to build state-space gating, which retains valid high-frequency and low-frequency components and suppresses fusion redundancy. Quantitative and qualitative experiments are conducted on public datasets, including Pavia University, Houston, and Chikusei, as well as the self-built nonhomologous real Huzhou dataset. Experimental results demonstrate that the proposed PMHIF-Net outperforms the other state-of-the-art methods and achieves satisfying generalization and robustness.**

***Index Terms*—Hyperspectral image (HSI), image fusion, Mamba, multispectral image (MSI), prior-guided.**

I. Introduction

Hyperspectral images (HSIs) are equipped with numerous contiguous spectral channels and offer superior spectral distinguishability, which can precisely depict the physical attributes of various terrestrial ground objects. Accordingly, HSIs have been widely applied in numerous remote sensing scenarios, including environmental monitoring [1], [2], precision agriculture [3], [4], and resource exploration [5], [6]. Nevertheless, limited by hardware imaging conditions, HSI sensors cannot achieve high spectral resolution and high-spatial-resolution simultaneously. In contrast, multispectral images (MSIs) possess superior spatial resolution. As a result, hyperspectral and MSI fusion (HMIF) technology, which fuses low-spatial-resolution HSI and high-resolution MSI, can compensate for the inherent information deficiencies of single data. It is capable of reconstructing high-quality HSIs with simultaneous high spatial and spectral resolution, and has become a research hotspot in the field of remote sensing image processing.

Existing HMIF methods can be roughly divided into two categories: traditional model-driven methods and deep learning (DL)-based data-driven methods. Traditional methods construct fusion frameworks based on manually designed prior constraints, and typical techniques include matrix decomposition [7], [8], tensor decomposition [9], [10], and Bayesian inference [11], [12]. Specifically, matrix decomposition formulates HSI as the product of spectral bases and spatial coefficients, and solves optimal solutions through low-rank constraints and regularization optimization. Tensor decomposition treats hyperspectral data as high-order tensors and decomposes them into multiple low-dimensional factors to preserve the inherent multidimensional structural features. Bayesian methods accomplish image reconstruction through probabilistic inference. Such methods feature simple principles, easy implementation, and clear physical interpretability, and do not rely on massive training datasets. However, they heavily depend on artificially defined prior rules, making it difficult to fully explore the deep nonlinear relationships in complex remote sensing data.

Benefiting from powerful nonlinear fitting and automatic feature learning capabilities, DL methods effectively overcome the defects of traditional strategies and have gradually become the mainstream solution for HMIF tasks. Convolutional neural networks (CNNs) can efficiently extract local features such as image edges and textures, and have been widely adopted in HMIF research [13], [14]. Multiple optimization strategies, including residual connection [15], multiscale modeling [16], attention mechanism [17], and improved loss functions [18], can further boost fusion performance. Even so, CNNs are

Received 3 August 2026; accepted 11 August 2026. Date of publication 14 August 2026; date of current version 26 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62277016 and Grant 42271409; in part by the "Pioneer" and "Leading Goose" Research and Development Program of Zhejiang Province, China, under Grant 2026C02A1239; and in part by Huzhou Municipal Science and Technology Bureau under Grant 2023GZ29. *(Corresponding author: Hui Zhang.)*

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Digital Object Identifier 10.1109/TGRS.2026.3723807

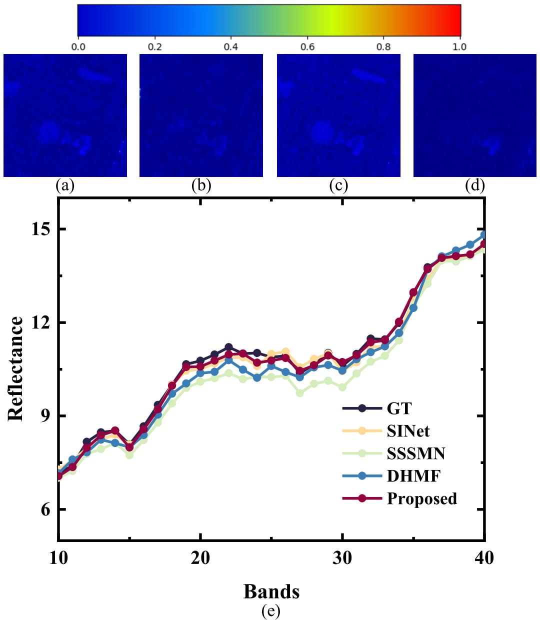

Fig. 1. Fusion error maps of (a) SINet, (b) SSSMN, (c) DHMF, and (d) proposed method. (e) Comparison of spectral reflectance curves for spectral gradient error evaluation.

restricted by fixed receptive fields and fail to capture long-range pixel correlations, leading to limited global modeling capacity. To tackle this problem, Transformer architectures have been introduced into HMIF tasks to establish global long-range dependencies via self-attention mechanisms [19], [20], [21]. However, its computational complexity increases quadratically with input size, resulting in excessive computing costs and restricting practical deployment efficiency. Built upon state-space models (SSMs), Mamba represents an innovative structure that captures global context under linear computation cost. It offers an efficient replacement for Transformers and has seen growing use in HMIF studies over recent years [22], [23], [24]. Current Mamba-based fusion methods leverage its outstanding long-range modeling ability to construct cross-modal interaction mechanisms, learn spatial–spectral feature correlations, and design lightweight fusion structures, achieving promising fusion results. Even so, these methods still suffer from three obvious limitations. First, spatial images contain directional edge structures that provide critical local information for feature learning. Most existing methods lack explicit perception of spatial edges in state-space modeling and cannot effectively distinguish edge regions from flat areas, thereby causing structural distortion and fusion errors, as displayed in Fig. 1(a)–(c). Second, spectral bands present gradual and sudden variations that reflect the dynamic change of ground object reflectance. Existing approaches fail to effectively capture the spectral change rate among adjacent bands, which tends to smooth spectral gradient details and induce severe spectral distortion, as shown in Fig. 1(e). Third, several methods adopt imperfect cross-modal interaction designs, leaving massive redundant information in the feature fusion process and degrading the final fusion accuracy.

Aiming at the aforementioned limitations, this study constructs a prior-guided Mamba hierarchical interactive fusion network, namely, PMHIF-Net. The overall architecture consists of four core components: spatial learning module (SpaLM), spectral learning module (SpeLM), spatial–spectral interactive Mamba block (SSIMB), and refined gated reconstruction module (RGRM). At the feature learning stage, SpaLM and SpeLM are elaborately designed to compensate for the insufficient spatial edge and spectral gradient perception of conventional Mamba. To solve the structural distortion problem caused by inadequate edge perception [Fig. 1(a)–(c)], this article embeds edge prior into SpaLM and constructs an edge-guided SSM (EGSSM). This design explicitly perceives image edge details and effectively reduces structural deviation of fusion results [Fig. 1(d)]. Aiming at the spectral distortion issue stemming from neglected interband spectral variation [Fig. 1(e)], SpeLM integrates a spectral gradient prior and builds a gradient-aware SSM (GASSM). It precisely captures the spectral reflectance variation tendency of adjacent bands, maintains spectral gradient consistency, and alleviates spectral distortion. At the feature interaction stage, SSIMB is developed to realize in-depth dual-modal feature interaction and generate multilevel progressive features through explicit cross-modal communication and dual-modal dynamic mutual control. At the feature reconstruction stage, RGRM couples multilevel features to enhance model robustness. Meanwhile, a frequency-domain state-space gate (FSSG) is introduced to embed frequency-domain prior information and separate and model high-frequency and low-frequency features. The adaptive feature screening mechanism retains valid detailed features and suppresses redundant information during fusion. The core contributions of the proposed method are outlined below.

1) SpaLM and SpeLM are designed to incorporate edge prior and spectral gradient prior into state-space modeling. SpaLM adopts dynamic edge gating to adaptively perceive multidirectional edge details, while SpeLM leverages a gradient-aware mechanism to characterize interband spectral variations and enhance the correlation learning of cross-channel features.
2) SSIMB is proposed to establish strong correlation interaction between spatial and spectral features. With multisource feature input, explicit cross-modal communication and dual-modal dynamic modulation, the proposed block improves the complementary utilization efficiency of multisource remote sensing information.
3) FSSG is designed to introduce a frequency-domain structural prior by separating high-frequency and low-frequency components. It adopts Mamba parallel modeling to generate frequency-domain adaptive spatial modulation weights, which effectively preserve fine high-frequency and low-frequency details and suppress redundant noise in feature reconstruction.
4) We construct an end-to-end layered fusion framework, PMHIF-Net, to boost stepwise perception and interaction of spatial–spectral features. Experiments on three open datasets reveal the superior performance of PMHIF-Net to the other state-of-the-art algorithms.

Further validations on a real nonhomogeneous dataset constructed from GF-5B and Sentinel-2A images verify the excellent cross-scene generalization ability and practical application value of PMHIF-Net.

II. Related Work

A. DL-Based HMIF Methods

As a typical data-driven learning paradigm, DL delivers powerful nonlinear modeling capability and end-to-end optimization superiority, which has made it the dominant technical solution for HMIF tasks. CNN serves as a vital tool for extracting local spectral and spatial features such as edges and textures due to its excellent local detail perception ability. Current CNN fusion frameworks split into two groups: model-guided fusion and fully data-driven fusion.

The former methods embed physical imaging constraints and prior knowledge into network optimization, improving the interpretability and stability of fusion results [25], [26], [27]. Huang et al. [28] developed DHIF-Net, which combines spatial–spectral iterative regularization with physical imaging models. The network adopts a spatially adaptive 3-D filter to capture spatial–spectral dependencies and formulates the fusion reconstruction process as a differentiable optimization problem, which closely conforms to actual imaging mechanisms.

Pure data-driven methods learn feature mapping rules directly from massive training data, making them adaptable to complex remote sensing scenarios [29], [30]. Hu et al. [17] designed HSRNet, which combines channel and spatial attention mechanisms with PixelShuffle upsampling strategies to achieve efficient spatial–spectral feature fusion and image reconstruction. Ran et al. [31] presented GuidedNet. With a high-resolution guidance branch and progressive reconstruction strategy, the method greatly reduces parameter overhead while maintaining high-precision image super-resolution and fusion performance. Although CNN can effectively extract local features, its fixed receptive field limits the capture of long-range pixel dependencies, leading to inherent deficiencies in global modeling [32].

To address the long-range modeling limitations of CNN, researchers have introduced Transformers into HMIF tasks. Transformers rely on self-attention to model global remote correlations and greatly lift fusion quality [33], [34], [35]. Hu et al. [19] created Fusformer, the first scheme introducing Transformer architectures into HMIF studies. He et al. [36] designed a local–global collaborative Transformer network (LGCT). The network adopts dual reverse feature streams for multiscale spatial–spectral feature extraction, leverages cross-window shuffling and grouped attention for global–local collaborative modeling, and optimizes reconstruction quality through hierarchical symmetric fusion. Liu et al. [37] developed SSTF-UNet, which embeds spatial and spectral Transformer modules into a U-shaped architecture to model global spatial and spectral dependencies separately. Xu et al. [38] built an enhanced spatial–frequency collaborative network (ESFS). By combining a channel spatial enhancement module (CSAM) and a selective frequency decomposition module (SFDM), the network jointly models global spatial dependencies and frequency-domain features. Liu et al. [39] designed a dual-stage spatial–spectral collaborative Transformer (DPFormer). With a dual-stage interaction mechanism and dual-stage Transformer residual block (DPTRB), DPFormer performs spectral injection and spatial enhancement in stages. Combined with a cross-scale integration block (CSIB) and frequency-domain perception loss, the method achieves high-precision HSI fusion. Xiao et al. [40] proposed the region-aware mixture-of-experts module (RAMoE) fusion network. They designed an attention sharing module (ASM) to facilitate cross-modal interaction. An RAMoE is adopted to adaptively focus on diverse ground-object regions and extract discriminative local features. A spatial–spectral reconstruction module (SSRM) is finally utilized to achieve high-fidelity image reconstruction. Nevertheless, the built-in self-attention module of Transformer yields quadratic computational overhead as input dimensions expand, which leads to heavy calculation burdens during model operation. Although existing studies adopt window constraint strategies [41], [42] to reduce computation, such approaches narrow the effective receptive field, weaken global dependency modeling ability, and still introduce redundant computational overhead.

B. State-Space Model

SSM was first introduced to DL with the proposal of the S4 model [43]. With the capability of global long-range modeling at linear computational complexity, SSM has evolved into an efficient modeling paradigm for DL tasks. On the basis of SSM, Gu and Dao [44] developed a Selective State-Space mechanism and further built the Mamba model. Such a design boosts sequence modeling flexibility and gains extensive usage across multiple research branches, such as language analysis, temporal sequence forecasting, and visual computing. According to diverse task requirements, existing studies have proposed various improved Mamba architectures. Peng et al. [45] established a bidirectional gated Mamba structure named PTM-Mamba to adapt to protein post-translational modification modeling tasks. Liu et al. [46] constructed a global context Mamba network (GCMNet) to handle long-time sequence forecasting. This model delivers precise prediction results through global patch embedding (GCPE) and dual variable-duration Mamba layers (VTM). To solve the restriction of unidirectional scanning in visual tasks, Liu et al. [47] developed VMamba equipped with 2-D selective scanning strategies, which can better adapt to the inherent feature extraction properties of image data.

Thanks to the strong global modeling capability, Mamba and its derivative models have gained growing popularity in HMIF. He et al. [48] pioneered Mamba application in pan-sharpening tasks and proposed Pan-Mamba. With specially designed channel exchange and cross-modal Mamba, the model realizes complementary enhancement of spatial and spectral features. Peng et al. [49] put forward FusionMamba with a dual U-Net structure, which adopts four-directional Mamba blocks and dedicated fusion blocks to achieve global perception and deep coupling of spatial and spectral features. Wu et al. [50] constructed a spectral–spatial cross Mamba network (SSCM), which utilizes spatial–spectral cross interaction to guide multiscale feature fusion. Xiao et al. [51] proposed the spatial invertible network SINet, which conducts parallel modeling via CNN and Mamba to adaptively fuse global and local features. However, most current HMIF methods that simply introduce Mamba or optimize scanning directions lack targeted modeling strategies for inherent heterogeneous characteristics of HSI and MSI, thereby leading to inadequate mining of spatial details and spectral signatures. In response to this limitation, several studies have designed independent Mamba branches for spatial and spectral dimensions. Feng et al. [52] proposed the sparse spatial–spectral Mamba network SSSMN. It adopts spatial–spectral attention to perform Top-k selection and preserve valid sequences to alleviate the forgetting issue of long-sequence information. Cui et al. [53] constructed the dual-stream hierarchical Mamba network DHMF-Net. A built-in multiscale attention scanning strategy is employed within spatial and spectral hierarchical Mamba blocks to strengthen the perception of multiscale structures. Nevertheless, the above Mamba-based methods fail to explicitly capture spatial edge details during state-space modeling. They also overlook critical features corresponding to the variation rate of spectral reflectance across bands. These limitations hinder the full exploitation of inherent modal properties and tend to cause structural distortion and spectral deviation. Besides, most existing methods lack effective guidance for bimodal interaction, leading to redundant information in feature fusion. To tackle these issues, this work embeds prior knowledge into the evolution of state spaces to explicitly guide SSM to capture spatial and spectral characteristics. Meanwhile, dynamic cross-modal interaction and frequency-aware feature refinement mechanisms are devised to mitigate fusion redundancy.

C. Comparison With Prior-Guided HMIF Methods

Prior knowledge leverages inherent physical laws in remote sensing imaging and imposes physical constraints on degradation processes, including spectral reflectance variation and geometric structural distortion. Pure data-driven models suffer obvious performance degradation when training samples are insufficient or scene distribution varies drastically. Combining prior knowledge with deep neural networks restricts network learning under real physical rules, compensating for the weak generalization and robustness of plain DL models and boosting the reliability and physical interpretability of fusion outputs.

Several existing HMIF methods integrate prior information into DL architectures. Zhang et al. [18] designed the spatial–spectral reconstruction network SSRNet. The framework calculates spatial and spectral edge discrepancies as gradient loss terms, which cooperate with reconstruction loss during joint optimization to retain fine edge textures. Dong et al. [54] proposed the model-guided deep super-resolution network MoG-DCN. This method embeds the imaging degradation model into network layers, builds a deep unfolding structure by unrolling iterative optimization algorithms, and introduces low-rank and sparsity priors of degradation models as regularizers in each iterative update. Liu et al. [55] put forward SEMF-Net, a multistage fusion network enhanced by spatial and spectral edges. It adopts edge prior operators for feature enhancement to strengthen edge responses, and feeds edge-boosted features as additional auxiliary inputs into subsequent network layers. All the above schemes integrate prior knowledge into CNN or Transformer backbones through indirect guidance strategies such as auxiliary loss functions, regularization penalties, and extra feature inputs. Different from these approaches, this work takes advantage of the unique property that priors can directly participate in state evolution within Mamba state-space modeling. Spatial edge and spectral gradient priors are embedded into state-space formulations, serving as internal control signals to directly govern the forward evolution of the whole network.

III. Methodology

A. State-Space Model

Fig. 2 illustrates the framework of PMHIF-Net, which consists of four core parts: SpaLM, SpeLM, SSIMB, and RGRM. In the feature learning stage, SpaLM focuses on mining spatial details and enhancing the perception of edge structures, while SpeLM models feature differences between spectral bands and strengthens spectral constraint capabilities. In the feature interaction stage, SSIMB realizes adaptive in-depth interaction and fusion of dual-modal features. In the feature reconstruction stage, RGRM performs multilevel feature aggregation and high-precision image reconstruction.

In this work, low-spatial-resolution hyperspectral data (LR-HSI) and high-spatial-resolution multispectral data (HR-MSI) are denoted as $X \in \mathbb{R}^{h\times w\times C}$ and $Y \in \mathbb{R}^{H\times W\times c}$, respectively. The target output high-resolution HSI (HR-HSI) is denoted as $Z \in \mathbb{R}^{H\times W\times C}$. Specifically, h and H ($H > h$) are the row numbers of input and target images, w and W ($W > w$) are the column numbers, and c and C ($C > c$) are the number of spectral bands. The core objective of the HMIF task is to fuse X and Y through PMHIF-Net to generate the target image Z with simultaneous high spatial and spectral resolution, which is defined as

$$Z = f(X, Y) \tag{1}$$

where $f(\cdot)$ is the overall mapping function of PMHIF-Net. The specific implementation pipeline of PMHIF-Net is detailed as follows.

We first apply 4× bicubic interpolation on LR-HSI X to generate upsampled hyperspectral features. X_{up} is concatenated with HR-MSI Y in the channel dimension, followed by a 3×3 convolution computing to adjust the channel dimension to C, yielding the shallow fused feature F_{cat}. This step completes the shallow correlation modeling of dual-modal data. The calculation process is expressed as

$$X_{up} = Upsample(X) \tag{2}$$

$$F_{cat} = Conv_1^3\left(Cat\left(X_{up}, Y\right)\right) \tag{3}$$

where $Upsample(\cdot)$ presents the bicubic interpolation upsampling operation. $Cat(\cdot)$ is channel-wise feature concatenation, and $Conv_1^3(\cdot)$ is standard convolution adopting a 3×3 kernel along with a unit stride.

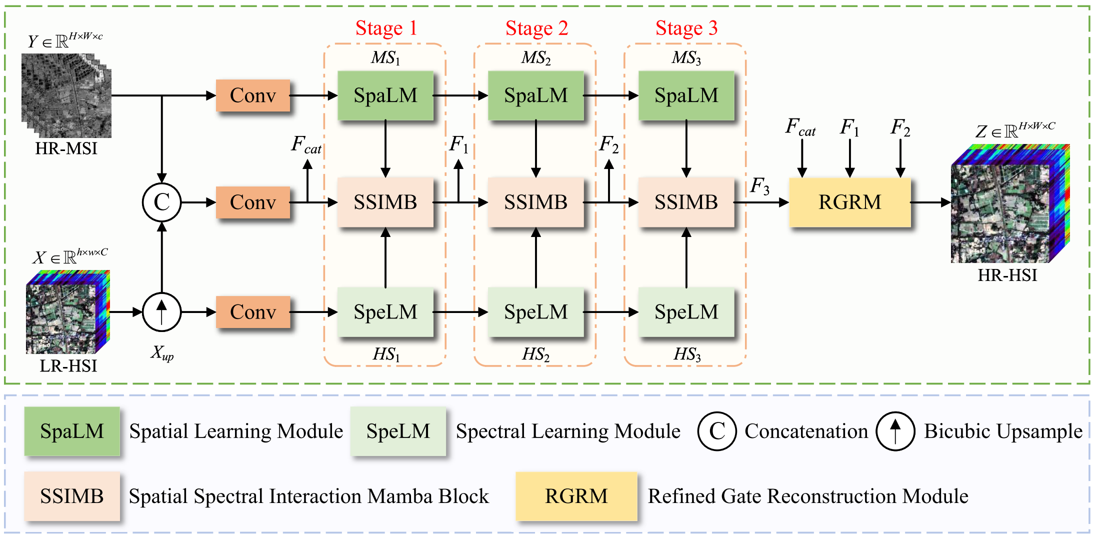

Fig. 2. Framework of PMHIF-Net.

Subsequently, X_{up} and Y are preprocessed by convolution and then fed into SpeLM and SpaLM, respectively, to extract initial spectral feature HS_1 and spatial feature MS_1. The obtained HS_1, MS_1, and F_{cat} are imported into SSIMB to complete the first cross-modal interaction, generating the multilevel interactive feature F_1. To achieve progressive deep feature learning, the network iteratively extracts features through SpeLM and SpaLM, sequentially acquiring HS_2, MS_2, HS_3, and MS_3. Meanwhile, these hierarchical features are successively sent to the subsequent SSIMB modules for progressive interaction, outputting F_2 and F_3 step by step. This design realizes the core progressive learning and interaction mechanism of spatial–spectral features. The corresponding formulas are given as follows:

$$HS_1 = SpeLM\left(Conv_1^3\left(X_{up}\right)\right) \tag{4}$$
$$HS_n = SpeLM\left(HS_{n-1}\right) \tag{5}$$
$$MS_1 = SpaLM\left(Conv_1^3\left(Y\right)\right) \tag{6}$$
$$MS_n = SpaLM\left(MS_{n-1}\right) \tag{7}$$
$$F_1 = SSIMB\left(HS_1, F_{cat}, MS_1\right) \tag{8}$$
$$F_n = SSIMB\left(HS_n, F_{n-1}, MS_n\right) \tag{9}$$

where $SpeLM(\cdot)$, $SpaLM(\cdot)$, and $SSIMB(\cdot)$ represent the feature mapping functions of SpeLM, SpaLM, and SSIMB, respectively.

Finally, the shallow fused feature F_{cat} and multilevel interactive features F_1, F_2, and F_3 are aggregated and input into RGRM for feature refinement and image reconstruction, producing the final fused image Z. The process is formulated as

$$Z = RGRM\left(F_{cat}, F_1, F_2, F_3\right) \tag{10}$$

where $RGRM(\cdot)$ denotes the mapping function of the RGRM.

B. Spatial Learning Module

SpaLM is designed to accurately capture global spatial dependencies of HR-MSI, strengthen the model's perception

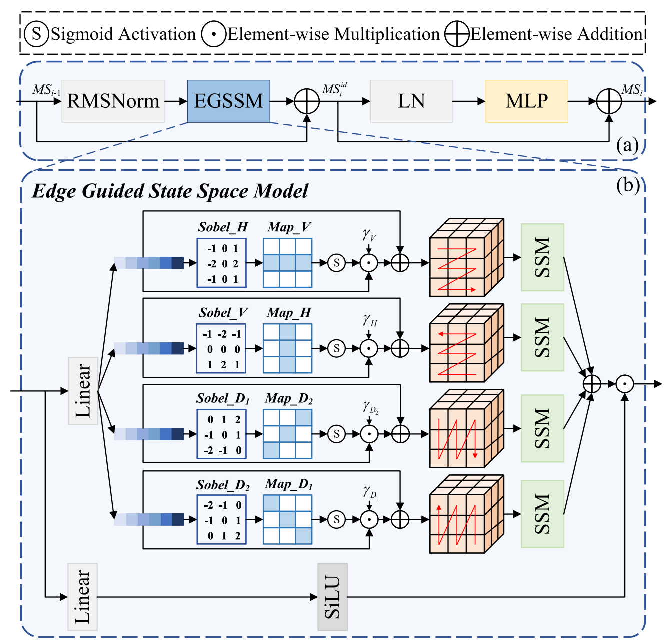

Fig. 3. Structural illustration of SpaLM.

of local details such as edges and textures, and realize progressive refinement learning of spatial features. Fig. 3(a) shows the internal layout of SpaLM. Centered on the EGSSM, this module performs global modeling on the input spatial feature MS_{i-1} to obtain intermediate residual feature MS_{i-1}^{id} with a residual connection. A multilayer perceptron (MLP) is further adopted to enhance the nonlinear representation capability, and the refined spatial feature MS_i is finally output. The overall calculation pipeline of SpaLM is formulated as follows:

$$MS_i^{id} = EGSSM\left(rms\left(MS_{i-1}\right)\right) + MS_{i-1} \tag{11}$$
$$MS_i = MLP\left(LN\left(MS_i^{id}\right)\right) + MS_i^{id} \tag{12}$$

where $rms(\cdot)$ denotes root mean square normalization, and $LN(\cdot)$ represents layer normalization operation.

Existing SS2D [47] realizes global image modeling through four-directional scanning. Nevertheless, its state update relies on implicit parameter learning, which fails to explicitly perceive high-frequency spatial structures such as image edges. It cannot effectively distinguish flat regions and mutation edge regions, thereby weakening the representation of key structural features and limiting the modeling accuracy of spatial details. To solve this problem, we develop EGSSM, which explicitly embeds edge prior into the parameter updating process of the state-space equation, and its structure is illustrated in Fig. 3(b).

For the input x, EGSSM adopts the Sobel operator to calculate edge feature maps in four directions, including horizontal (H), vertical (V), and two diagonal directions (D1 and D2), so as to generate edge response $edge_n$. Edges in different directions present anisotropic structural differences in physical characteristics. Therefore, an independent learnable gate parameter w is configured for each direction. The sigmoid function activates these parameters to generate modulation intensity weight γ, which is weighted and fused with the corresponding edge weight to construct a dynamic edge gating mechanism. This mechanism guides the state-space equation updating of each scanning sequence, strengthens feature response in edge regions, and finally yields edge-enhanced feature x'. With this design, each scanning direction can adaptively capture corresponding edge structures, and the SSM can enhance modeling performance at high-frequency mutation edges and textures. The 2DSSM establishes long-range dependencies in four directions separately, and all enhanced directional features are fused to complete global spatial modeling with reinforced local edge perception to obtain x_e. The specific calculation procedures are expressed as

$$\begin{aligned} &Map_V, Map_H, Map_{D_1}, Map_{D_2} \\ &\quad = |Sobel_n(x)|, \quad n = \{H, V, D_2, D_1\} \end{aligned} \tag{13}$$

$$edge_n = \sigma(Map_n), \quad n = \{H, V, D_2, D_1\} \tag{14}$$

$$\gamma_n = \sigma(w_n), \quad n = \{H, V, D_2, D_1\} \tag{15}$$

$$x'_n = x \odot (1 + \gamma_n \odot edge_n), \quad n = \{H, V, D_2, D_1\} \tag{16}$$

$$x_e = \sum_{i=1}^{4} EGSS2D\left(x'_i\right) \tag{17}$$

where $Sobel(\cdot)$ and $Map(\cdot)$ represent the Sobel matrix and the edge feature map in the corresponding direction. w_n and γ_n represent the learnable parameter and its normalized learning weight for directional modulation. $edge_n$ represent the edge response weight, and σ_n represent the Sigmoid activation function. $EGSS2D(\cdot)$ represent the edge-guided 2DSSM operation. Algorithm 1 illustrates the single time-step state updating procedure of EGSSM in detail.

C. Spectral Learning Module

To effectively explore global spectral correlations across channels of LR-HSI, capture subtle feature differences between adjacent bands, and enable the network to progressively refine spectral perception during iterative learning, this work develops SpeLM. The structure of SpeLM is displayed in Fig. 4(a). Centered on the GASSM, this module imposes

Algorithm 1 Edge-Guided State-Space Update

Input: x_t, h_{t-1}
Output: y_t, h_t

$\tilde{x}_t \leftarrow x_t \cdot (1 + \gamma_n \cdot \text{edge}_n)$
/* edge-gated input modulation */
$\Delta(\tilde{x}_t), \mathbf{B}(\tilde{x}_t), \mathbf{C}(\tilde{x}_t) \leftarrow \text{Linear}(\tilde{x}_t)$
/* Δ: time step, $\mathbf{B}, \mathbf{C}$: projection matrices */
$\bar{\mathbf{A}}_t \leftarrow \exp(\Delta(\tilde{x}_t) \cdot \mathbf{A})$
/* $\mathbf{A}$: state transition matrix, $\bar{\mathbf{A}}_t$: discretized A */
$\bar{\mathbf{B}}_t \leftarrow (\Delta \cdot \mathbf{A})^{-1} \cdot (\exp(\Delta(\tilde{x}_t) \cdot \mathbf{A}) - \mathbf{I}) \cdot \Delta(\tilde{x}_t) \cdot \mathbf{B}(\tilde{x}_t)$
/* $\bar{\mathbf{B}}_t$: discretized B, $\mathbf{I}$: identity matrix */
$h_t \leftarrow \bar{\mathbf{A}}_t \cdot h_{t-1} + \bar{\mathbf{B}}_t \cdot \tilde{x}_t$
/* h_t: hidden state */
$y_t \leftarrow \mathbf{C}(\tilde{x}_t) \cdot h_t + \mathbf{D} \cdot \tilde{x}_t$
/* $\mathbf{D}$: skip connection */
return y_t, h_t

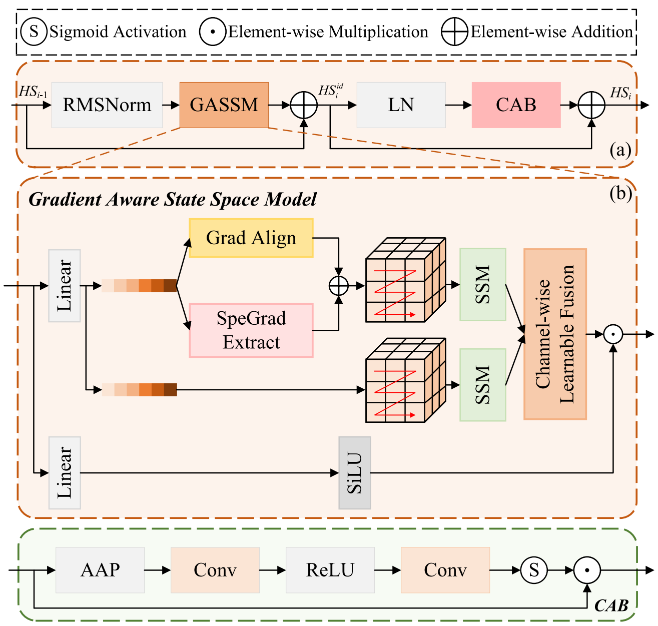

Fig. 4. Structural illustration of SpeLM.

spectral gradient perception on input feature HS_{i-1} to mine nonlinear band-wise correlations. Intermediate residual feature HS_{i-1}^{id} is obtained with residual connections, and a channel attention block (CAB) is subsequently adopted to adaptively enhance spectral awareness, yielding the final refined output feature HS_i. This scheme substantially enhances the model's spectral characterization performance. The overall processing pipeline of SpeLM is formulated as follows:

$$HS_i^{id} = GASSM(rms(HS_{i-1})) + HS_{i-1} \tag{18}$$

$$HS_i = CAB\left(LN\left(HS_i^{id}\right)\right) + HS_i^{id} \tag{19}$$

where $GASSM(\cdot)$ represents the gradient-aware state-space modeling operation, and $CAB(\cdot)$ represents the channel attention enhancement process.

The spectral curves of HSI exhibit complex nonlinear gradual changes and abrupt variations. Traditional Mamba-based spectral modeling only learns band correlations in an implicit manner, making it difficult to characterize the dynamic variation rules of reflectance among spectral bands. In contrast, the first-order spectral gradient can intuitively reflect the variation rate of adjacent bands and precisely locate the position and intensity of ground object absorption characteristics. On this basis, GASSM is designed in this study, and its structure is shown in Fig. 4(b).

GASSM first calculates the first-order spectral gradient of input feature y along the channel dimension to acquire band variation information. A learnable linear transformation is applied to achieve spatial alignment and adaptive fusion of the original spectral feature and gradient feature, generating the gradient-enhanced feature g. A dual-branch independent SSM modeling architecture is then constructed to perform global spectral modeling on the original feature y and gradient-enhanced feature g, respectively. To realize gradient-guided optimization, a channel-wise learnable fusion mechanism is introduced. Learnable parameter θ is assigned to each spectral band for independent weight allocation, while parameters are shared in the spatial dimension to maintain physical consistency of spectral information and avoid redundant spatial differences. The dual-branch feature y_e is finally fused via weighted aggregation to implement refined spectral modeling guided by a gradient prior, which effectively strengthens the model's ability to perceive and distinguish subtle spectral variations. The specific calculations are given as follows:

$$g = SpeGE(y) + \phi(y) \tag{20}$$
$$\lambda = \sigma(\theta) \tag{21}$$
$$y_e = \lambda \odot SSM(y) + (1-\lambda) \odot SSM(g) \tag{22}$$

where $SpeGE(\cdot)$ denotes channel-wise gradient extraction. $\phi(\cdot)$ denotes the gradient alignment operation. λ refers to the learnable fusion weight, and $SSM(\cdot)$ is the state-space modeling operator.

D. Spectral–Spatial Interaction Mamba Block

To achieve correlated guided interaction and adaptive fusion between spatial and spectral information, as well as to support hierarchical progressive enhancement within the network, this article develops SSIMB. Equipped with explicit cross-modal communication and dynamic adaptive fusion strategies, SSIMB enables mutual constraints between spatial and spectral streams during state-space modeling and retains valid feature information. Its detailed structure is visualized in Fig. 5.

The module first implements explicit bidirectional mutually controlled interactive modeling for input spatial feature MS_i and spectral feature HS_i. Specifically, linear projections first generate state-space branches and learnable gate branches from inputs. Within each state-space branch, depthwise convolution enriches local neighborhood information prior to SSM modeling. Bidirectional mutual control guidance is then carried out: weights from the spectral branch regulate spatial feature modeling, while weights from the spatial branch constrain spectral feature learning. Finally, the two interacting features are superimposed to obtain the cross-modal coupling

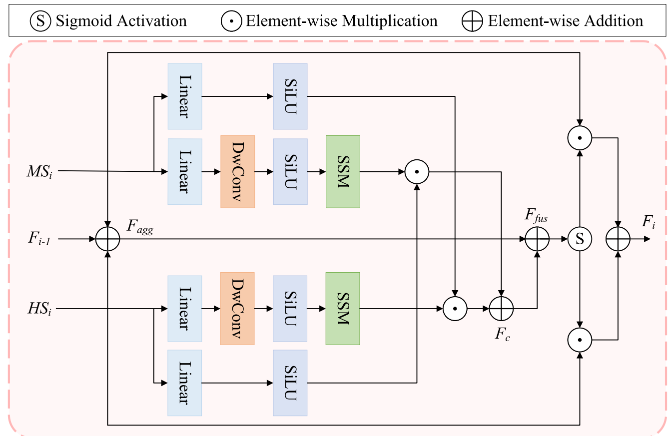

Fig. 5. Structural illustration of SSIMB.

feature, achieving mutual guidance and collaborative optimization of heterogeneous modalities. This process can be expressed as

$$f_{ssm} = SSM\left(SiLU\left(DwConv_1^3\left(Linear(\cdot)\right)\right)\right) \tag{23}$$
$$f_g = SiLU\left(Linear(\cdot)\right) \tag{24}$$
$$F_c = f_{ssm}(MS_i) \odot f_g(HS_i) + f_{ssm}(HS_i) \odot f_g(MS_i) \tag{25}$$

where $Linear(\cdot)$ denotes linear projection operation. $SiLU(\cdot)$ denotes SiLU activation function, and $DwConv_1^3(\cdot)$ refers to depthwise convolution adopting a 3×3 kernel along with a unit stride.

Subsequently, the current-level spatial feature MS_i, spectral feature HS_i, and previous-layer interactive feature F_i are summed to construct shallow coupling feature F_{agg} containing multilevel spatial–spectral information. Then, F_{agg} is further fused with the cross-modal coupling feature F_c generated by bidirectional mutual control to produce preliminary deep interactive feature F_{fus}. To further strengthen valid spatial–spectral correlations and eliminate redundant fusion information, this work adopts an adaptive fusion mechanism. It adaptively generates modulation weight from MS_i and HS_i, and implements adaptive weighted calibration on the preliminary fused feature F_{fus} to obtain the refined interactive feature F_i at the current layer. This design enables deep coupling of cross-modal features and effectively filters out redundant information during feature fusion. The detailed calculation process is formulated as

$$F_{agg} = MS_i + F_{i-1} + HS_i \tag{26}$$
$$F_{fus} = F_{agg} + F_c \tag{27}$$
$$F_i = MS_i \odot \sigma\left(F_{fus}\right) \odot F_{fus} + HS_i \odot \left(1 - \sigma\left(F_{fus}\right)\right) \odot F_{fus}. \tag{28}$$

This work adopts a multilevel progressive strategy to sustain continuous correlation transmission of spatial and spectral features throughout the network. Deep-layer features are iteratively refined on the basis of shallow counterparts, yielding multigranularity hierarchical features for subsequent reconstruction.

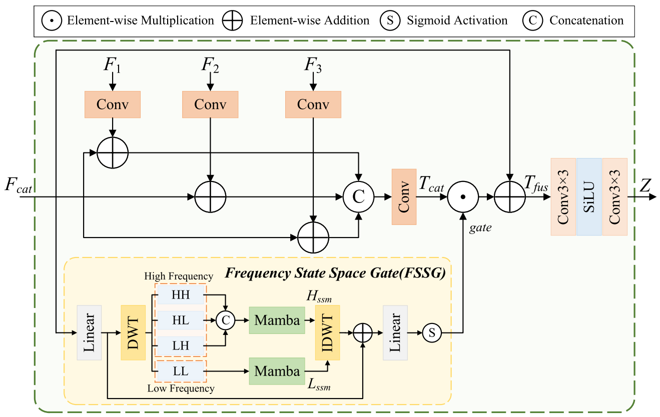

Fig. 6. Structural illustration of RGRM.

E. Refined Gated Reconstruction Module (RGRM)

We build RGRM to dynamically calibrate weights of hierarchical features, strengthen model stability, filter useless signals, and retain fine details in recovered images, as visualized in Fig. 6. Hierarchical interactive features F_1, F_2, and F_3 are processed by convolution, and then superposed with the shallow fused feature F_{cat}, respectively, to realize information complementarity between deep and shallow features. We stack the three refined feature groups across channels, then use convolution to align the channel count to the target C and get the fused feature T_{cat}. Second, the shallow fused feature T_{cat} is fed into the frequency state-space gate (FSSG) for frequency-domain-aware feature screening. The obtained global adaptive gate weight $gate$ performs element-wise multiplication with the aggregated feature T_{cat}. Combined with the residual connection of F_{cat} to preserve original valid features, the refined fusion feature T_{fus} is acquired. Finally, T_{fus} is fed into a multistage convolution block for feature reconstruction, and the ultimate fused image Z is output. The overall formulations of RGRM are summarized as follows:

$$T_{cat} = Conv_1^3\Big(Cat\Big(\big(Conv_1^3(F_1) + F_{cat}\big),\ \big(Conv_1^3(F_2) + F_{cat}\big), \big(Conv_1^3(F_3) + F_{cat}\big)\Big)\Big) \tag{29}$$

$$gate = FSSG(F_{cat}) \tag{30}$$

$$T_{fus} = gate \odot T_{cat} + F_{cat} \tag{31}$$

$$Z = Conv_1^3\left(SiLU\left(Conv_1^3\left(T_{fus}\right)\right)\right) \tag{32}$$

where $FSSG(\cdot)$ represents the frequency domain state-space gated operation, and $gate$ represents the gating weight.

FSSG targets redundant suppression via global perception of frequency-domain structures. Specifically, although Mamba supports effective feature screening in global modeling, it lacks discrimination for diverse feature structures such as edge textures, global structural backgrounds, and noise artifacts. In remote sensing images, high-frequency components contain abundant local edge and texture details, while low-frequency components maintain global structural semantics. Frequency-domain decomposition explicitly separates low-frequency and high-frequency contents. When combined with Mamba, this decomposition enables effective perception of primary structures embedded in low-frequency bands and edge details carried by high-frequency bands. Accordingly, FSSG employs discrete wavelet transform (DWT) to decouple input features into low-frequency components and multidirectional high-frequency components. Two parallel Mamba branches conduct global modeling for high-frequency features H_{ssm} and low-frequency features L_{ssm}. Inverse DWT reconstructs frequency-domain features. An activation function finally generates adaptive gating weights to select and preserve valid high-frequency and low-frequency details as well as suppress redundant information. The specific calculation procedures are given as

$$HH, HL, LH, LL = DWT(Linear(F_{cat})) \tag{33}$$

$$H_{ssm} = Mamba(Cat(HH, HL, LH)) \tag{34}$$

$$L_{ssm} = Mamba(LL) \tag{35}$$

$$gate = \sigma\Big(Linear\Big(IDWT(H_{ssm}, L_{ssm}) + Linear(F_{cat})\Big)\Big) \tag{36}$$

where HH, HL, and LH indicate high-frequency components in diagonal, horizontal, and vertical directions, respectively, and LL denotes the low-frequency component. $DWT(\cdot)$ and $IDWT(\cdot)$ represent DWT and inverse DWT operations.

IV. Experiments and Analysis

A. Datasets

We test PMHIF-Net's fusion effect and generalization over three open hyperspectral datasets (PaviaU, Houston, and Chikusei) plus a self-built Huzhou real dataset combining GF-5B and Sentinel-2A data. These datasets differ in spectral coverage, spatial resolution, and surface land types, suitable for assessing model performance across diverse data distributions and practical, complicated scenes. Detailed descriptions of each dataset are provided in the following.

Carried by the ROSIS airborne spectrometer, the PaviaU dataset covers the campus zone of Pavia University in Italy. This urban dataset records various ground objects blending artificial constructions and natural vegetation. Houston dataset originates from ITRES CASI-1500 spectrometer surveys across Houston region of America. Its abundant land cover categories consist of architecture, traffic roads, plants, and water surfaces. The Chikusei dataset originates from airborne headwall hyperspec-VNIR-C scans above Chikusei, Japan. This high-resolution data captures mixed urban and rural terrain covered with croplands, constructions, and various vegetation. Due to the presence of black invalid shadow areas at the edges of the original images, this study selected the effective areas with rich features in the center of the images for the experiments.

Huzhou dataset is a novel real cross-source dataset developed in this work, which contains matched hyperspectral and MSI pairs. HSIs of this dataset are captured by the AHSI sensor carried by the GF-5B satellite, covering the southwest area of Changxing County, Huzhou, on January 11, 2023. The MSI data is obtained through homologous matching by the Sentinel-2A satellite on January 8, 2023. The scene is mainly composed of large areas of farmland, with a small number

TABLE I
DETAILED PARAMETERS OF ALL EXPERIMENTAL DATASETS

Datasets	Simulated Scenes			Real Scenes(HuZhou)	
	PaviaU	Houston	Chikusei	GF-5B(HSI)	Sentinel-2A(MSI)
Numbers of bands	103	144	128	106	4
Pixel resolution(m)	1.3	2.5	2.5	30	10
Spectral range(nm)	430–860	364–1046	343–1018	455–908	492–832
Size($H \times W$)	610×340	349×1905	1024×512	1800×1800	600×600

of buildings as supplementary. Before experiments, radiometric calibration, atmospheric correction, orthorectification, and precise spatial registration are uniformly implemented for all images using ENVI software. The bands in the HSI that match the spectral coverage of the MSI are selected for the subsequent experiments. The detailed parameters of each dataset are given in Table I.

B. Implementation Details

All the experiments strictly follow the Wald protocol [56] to construct the datasets and experimental paradigms. Experimental configurations fall into two categories: simulated data and real-scene data. We process the three public simulated datasets as follows: original high-resolution hyperspectral data are blurred via a 5×5 kernel with a standard deviation of 2, then downscaled by a factor of 4 to create low-resolution hyperspectral inputs. Meanwhile, five feature bands of the original HR-HSI are sampled at equal intervals to generate HR-MSI data. To prevent data leakage, we extract a central 128×128 HR-HSI patch for testing and validation, with all leftover regions reserved for training. This partition eliminates spatial overlap between training and test samples. In training, random cropping is applied on training-region images to produce network inputs: $32 \times 32 \times C$ LR-HSI and $128 \times 128 \times 5$ HR-MSI patches.

For the real Huzhou dataset without available ground-truth 10-m HR-HSI, this work adopts a downsampling progressive training strategy; the original 30-m GF-5B HSI is treated as the reference ground truth (GT). Gaussian blurring with a 3×3 kernel and standard deviation of 0.5, together with threefold downsampling, is adopted to generate degraded data, forming training pairs consisting of 90-m LR-HSI and 30-m MSI. In each training iteration, image patches with sizes of $48 \times 48 \times 106$ for LR-HSI and $144 \times 144 \times 4$ for MSI are randomly cropped from spatially matched regions. After completing the degradation-based training, the well-trained model weights are adopted for full-resolution inference. During testing, the network receives real 30-m LR-HSI and 10-m HR-MSI as inputs and outputs reconstructed 10-m HR-HSI. Test patches are cropped from spatially coincident regions with dimensions of $48 \times 48 \times 106$ for 30-m HSI and $144 \times 144 \times 4$ for 10-m MSI.

All experiments were implemented based on the PyTorch framework. The hardware configuration is an Intel Core Ultra 9 285K (3.70 GHz) CPU and an NVIDIA GeForce RTX 5090D GPU (24 G). The network is optimized with the mse loss function and trained iteratively using the Adam optimizer. The cosine annealing strategy is adopted for learning rate scheduling. The initial learning rate is set to 1×10^{-4} and the minimum learning rate is 1×10^{-5}. The batch size is set to 1, and the training process runs for 20 000 epochs. No data augmentation techniques are utilized in the experiments.

C. Comparison Methods and Evaluation Metrics

To fully verify the performance advantages of PMHIF-Net, this study conducts a comparative experiment by selecting nine state-of-the-art fusion algorithms. These methods cover three mainstream technical frameworks, including CNN-based MoGDCN [54], DHIF [28], and HSRNet [17]; Transformer-based LGCT [36], SSTF-UNet [37], and ESFS [38]; and Mamba-based SINet [51], SSSMN [52], and DHMF [53]. All competing approaches are retuned under the unified experimental configurations described in Section IV-B using their default hyperparameters and fully optimized to reach peak performance for impartial comparison.

This article comprehensively assesses the fusion performance using mainstream quantitative evaluation indicators. For the simulated datasets with real References, four types of indicators are selected, namely, peak signal-to-noise ratio (PSNR), structural similarity (SSIM), spectral angle mapper (SAM), and relative dimensionless global error in synthesis (ERGAS). PSNR and SSIM are used to quantify the spatial detail restoration degree, with higher values indicating better spatial fidelity; SAM and ERGAS quantify spectral distortion and overall fusion quality, and smaller values correspond to more accurate fusion outcomes. Since GT is unavailable for real remote sensing data, we adopt the no-reference metric QNR [57] for assessment. This metric weights spatial distortion D_λ and spectral distortion D_s for joint evaluation. D_λ and D_s approach zero when distortion weakens, while QNR reaching unity represents superior fused imagery.

D. Experimental Results and Analysis on Simulated Datasets

Table II lists quantitative metrics of all contrast algorithms across three simulated datasets. Our PMHIF-Net outperforms other competitors on every evaluation metric. On the PaviaU dataset, the PSNR of PMHIF-Net is 0.16 dB higher than that of the method achieving the suboptimal value, while ERGAS and SAM are reduced by 2.0% and 1.6%, respectively, and DHMF achieves the suboptimal SSIM, indicating its good structure preservation ability. On Houston dataset, the PSNR of PMHIF-Net is 0.60 dB higher than that of the method achieving the suboptimal value, while ERGAS and SAM are reduced by 3.6% and 7.1%, respectively, and LGCT and

TABLE II
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON SIMULATED DATASETS (BOLD FONTS DENOTE THE OPTIMAL VALUES, AND UNDERLINED FONTS DENOTE THE SUBOPTIMAL VALUES)

Methods	PaviaU				Houston				Chikusei			
	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM
Ideal value	$+\infty$	0	0	1	$+\infty$	0	0	1	$+\infty$	0	0	1
MoGDCN	43.1061	1.2882	2.0101	0.9793	48.2062	1.2793	1.7067	0.9899	40.1690	2.0577	1.4237	0.9709
DHIF	43.5284	1.2054	1.9467	0.9812	48.9154	1.2095	1.5005	0.9900	41.8258	1.8198	1.3436	0.9726
HSRNet	43.3317	1.2313	1.9819	0.9812	47.7160	1.4605	1.7434	0.9878	41.2918	1.8022	1.4071	0.9738
LGCT	44.5763	1.1154	1.7442	0.9825	51.2918	<u>1.0161</u>	1.2156	<u>0.9920</u>	<u>43.4825</u>	1.6690	<u>1.0939</u>	0.9750
SSTF-UNet	44.1499	1.1580	1.8313	0.9813	49.1525	1.1691	1.4523	0.9904	43.0928	1.8287	1.1330	0.9722
ESFS	44.3903	1.1283	1.7772	0.9826	50.7540	1.1093	1.2586	0.9915	43.1730	1.5825	1.1064	<u>0.9778</u>
SINet	44.2602	1.1386	1.8093	0.9822	49.8765	1.1608	1.3346	0.9904	42.7030	1.8689	1.2000	0.9735
SSSMN	43.6848	1.2113	1.8767	0.9812	50.3988	1.0619	1.2320	<u>0.9920</u>	41.5832	1.9682	1.2373	0.9686
DHMF	<u>44.6536</u>	<u>1.1087</u>	<u>1.7422</u>	<u>0.9828</u>	<u>51.5080</u>	1.0190	<u>1.1602</u>	0.9919	43.3834	<u>1.5440</u>	1.1039	0.9771
PMHIF-Net	**44.8173**	**1.0860**	**1.7136**	**0.9831**	**52.1071**	**0.9796**	**1.0777**	**0.9930**	**44.0506**	**1.4041**	**1.0514**	**0.9806**

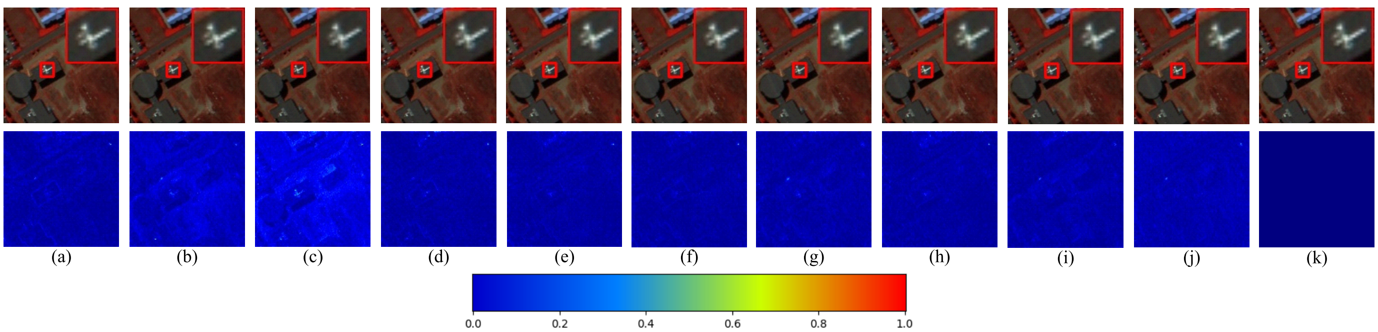

Fig. 7. Qualitative fusion contrast for the PaviaU dataset. The top row presents pseudocolor fused imagery (R-81, G-41, and B-21), and the bottom row displays residual maps against reference images. (a) MoGDCN, (b) DHIF, (c) HSRNet, (d) LGCT, (e) SSTF-UNet, (f) ESFS, (g) SINet, (h) SSSMN, (i) DHMF, (j) PMHIF-Net, and (k) GT.

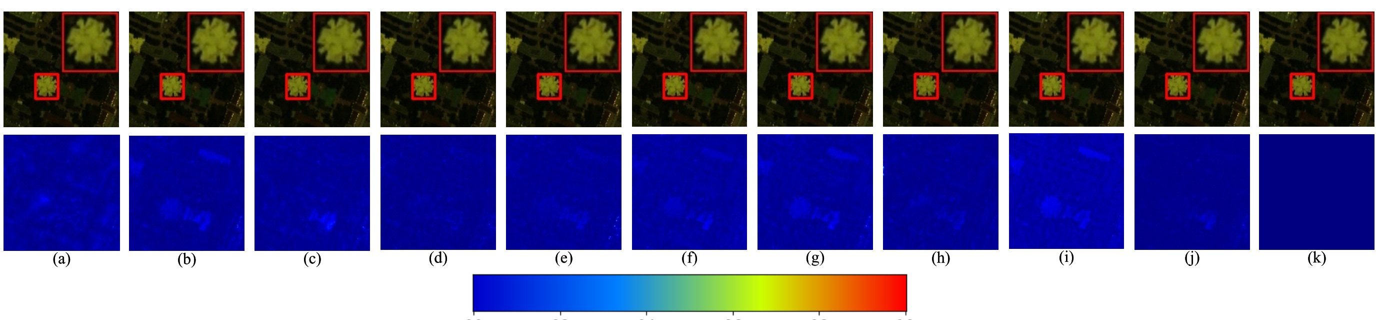

Fig. 8. Qualitative fusion contrast for Houston dataset. The top row presents pseudocolor fused imagery (R-61, G-30, and B-8), and the bottom row displays residual maps against reference images. (a) MoGDCN, (b) DHIF, (c) HSRNet, (d) LGCT, (e) SSTF-UNet, (f) ESFS, (g) SINet, (h) SSSMN, (i) DHMF, (j) PMHIF-Net, and (k) GT.

SSSMN simultaneously achieve the suboptimal SSIM results. On the Chkusei dataset, the PSNR of PMHIF-Net is 0.57 dB higher than that of the method achieving the suboptimal value, while ERGAS and SAM are reduced by 9.1% and 3.9%, respectively, and ESFS is the suboptimal SSIM method. Horizontal comparison of different algorithm architectures reveals that global modeling methods based on Transformer and Mamba consistently outperform conventional CNN-based approaches. This phenomenon validates the significance of long-range dependency modeling for HMIF tasks. The superior performance of PMHIF-Net benefits from the prior-guided dual-modal progressive learning mechanism. Edge prior and spectral gradient prior, respectively, strengthen spatial detail perception and interband variation capture. Sufficient cross-modal interaction promotes complementary learning of spatial and spectral features, while the frequency-domain prior gating mechanism filters valid features and suppresses fusion redundancy, ultimately achieving high-precision hyperspectral fusion results.

Figs. 7–9 display pseudocolor fusion outputs from all algorithms alongside their respective error distribution maps. The visualization results indicate that PMHIF-Net produces fusion results with the best spatial structure and spectral fidelity on all three simulated datasets, which are the most consistent

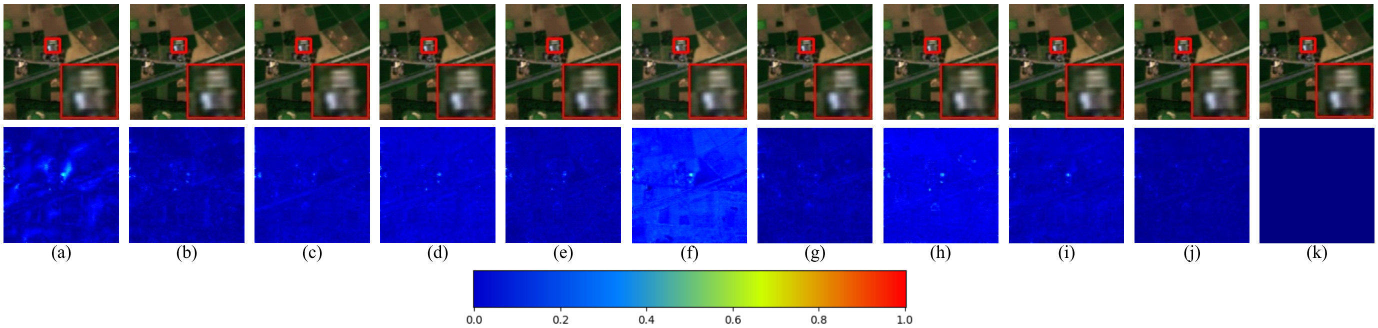

Fig. 9. Qualitative fusion contrast for Chikusei dataset. The top row presents pseudocolor fused imagery (R-60, G-40, and B-20), and the bottom row displays residual maps against reference images. (a) MoGDCN, (b) DHIF, (c) HSRNet, (d) LGCT, (e) SSTF-UNet, (f) ESFS, (g) SINet, (h) SSSMN, (i) DHMF, (j) PMHIF-Net, and (k) GT.

TABLE III

Quantitative Evaluation Results of Downscaled and Full-Scale Experiments on the Real Huzhou Dataset (Bold Fonts Denote Optimal Values, and Underlined Fonts Denote Suboptimal Values)

Methods	Reduced-Resolution				Full-Resolution		
	PSNR	ERGAS	SAM	SSIM	QNR	D_λ	D_s
Ideal value	$+\infty$	0	0	1	1	0	0
MoGDCN	32.2799	2.0771	1.8488	0.9130	0.9233	0.0213	0.0566
DHIF	34.7317	1.6852	1.6965	0.9448	0.9197	0.0403	0.0416
HSRNet	36.0392	1.4793	1.4960	0.9612	0.9095	0.0531	0.0395
LGCT	32.8636	1.8650	1.8650	0.9282	0.9339	0.0214	0.0457
SSTF-UNet	36.5442	1.3482	1.2890	0.9605	0.9496	0.0162	0.0348
ESFS	35.6460	1.5413	1.5773	0.9566	0.9326	0.0359	0.0327
SINet	33.8068	1.8503	1.5791	0.9337	0.9384	0.0183	0.0441
SSSMN	33.5080	1.8294	1.5403	0.9314	0.9211	0.0206	0.0592
DHMF	35.6124	1.4865	1.4461	0.9570	0.9439	0.0244	0.0325
PMHIF-Net	**36.8043**	**1.2808**	**1.2483**	**0.9651**	**0.9541**	**0.0157**	**0.0307**

with the GT. On the PaviaU dataset (Fig. 7), DHIF and HSRNet show obvious deviations and large reconstruction errors, whereas PMHIF-Net presents minimal overall errors and superior spatial detail restoration. On Houston dataset (Fig. 8), several methods, including DHIF, HSRNet, ESFS, SINet, and DHMF, suffer from severe distortion and artifacts in multiple regions. Although LGCT and SSSMN achieve relatively better visual performance, their detail preservation is still inferior to PMHIF-Net. The error map of PMHIF-Net exhibits the highest consistency with the GT. On the Chikusei dataset (Fig. 9), most comparison methods have varying degrees of bright spot artifacts, among which MoGDCN shows large areas of distortion deviation, while PMHIF-Net has no obvious bright spots or distortion problems, and the overall fusion effect has the highest consistency with the GT. The visualization results fully prove that PMHIF-Net can effectively achieve the collaborative learning of local edge details and global features. Relying on precise cross-modal interaction and feature refinement mechanisms, it can completely retain the critical spatial structure and spectral features, and significantly improve the overall fusion quality.

To further verify the spectral fidelity of PMHIF-Net, spectral reflectance comparison curves are plotted at three distinct pixel coordinates for each simulated dataset, as illustrated in Fig. 10. The spectral comparisons reveal evident discrepancies in spectral preservation capacity among different methods across all datasets. For the PaviaU dataset, Fig. 10(b) demonstrates that all competing methods exhibit obvious deviations within bands 15–25 at coordinate (60, 6). Benefiting from the explicit perception of first-order spectral gradients during state-space modeling, PMHIF-Net closely fits the sharp descending and ascending trends of the ground-truth curve and achieves prominent advantages in spectral fidelity. For Houston dataset, the curve comparison at coordinate (64, 64) in Fig. 10(d) shows severe spectral distortion in HSRNet, whose curve greatly deviates from the GT over bands 17–30. Although other competitors roughly follow the trend of the GT, the proposed PMHIF-Net achieves the highest consistency with the GT at steep spectral regions for pixels corresponding to three different land cover types on this dataset, without spectral distortion or offset. For the Chikusei dataset, DHIF produces substantial curve deviations over bands 32–50 at coordinate (50, 50), as shown in Fig. 10(g), while MoGDCN presents overall biases at coordinate (20, 66) in Fig. 10(h). When dealing with diverse land cover types, PMHIF-Net maintains minimal errors relative to GT throughout all bands and delivers stable band fitting performance. These results sufficiently validate that GASSM enables PMHIF-Net to explicitly model spectral gradient information. The network effectively captures abrupt and gradual variations across adjacent spectral bands for various ground objects and attains superior spectral fidelity in hyperspectral reconstruction.

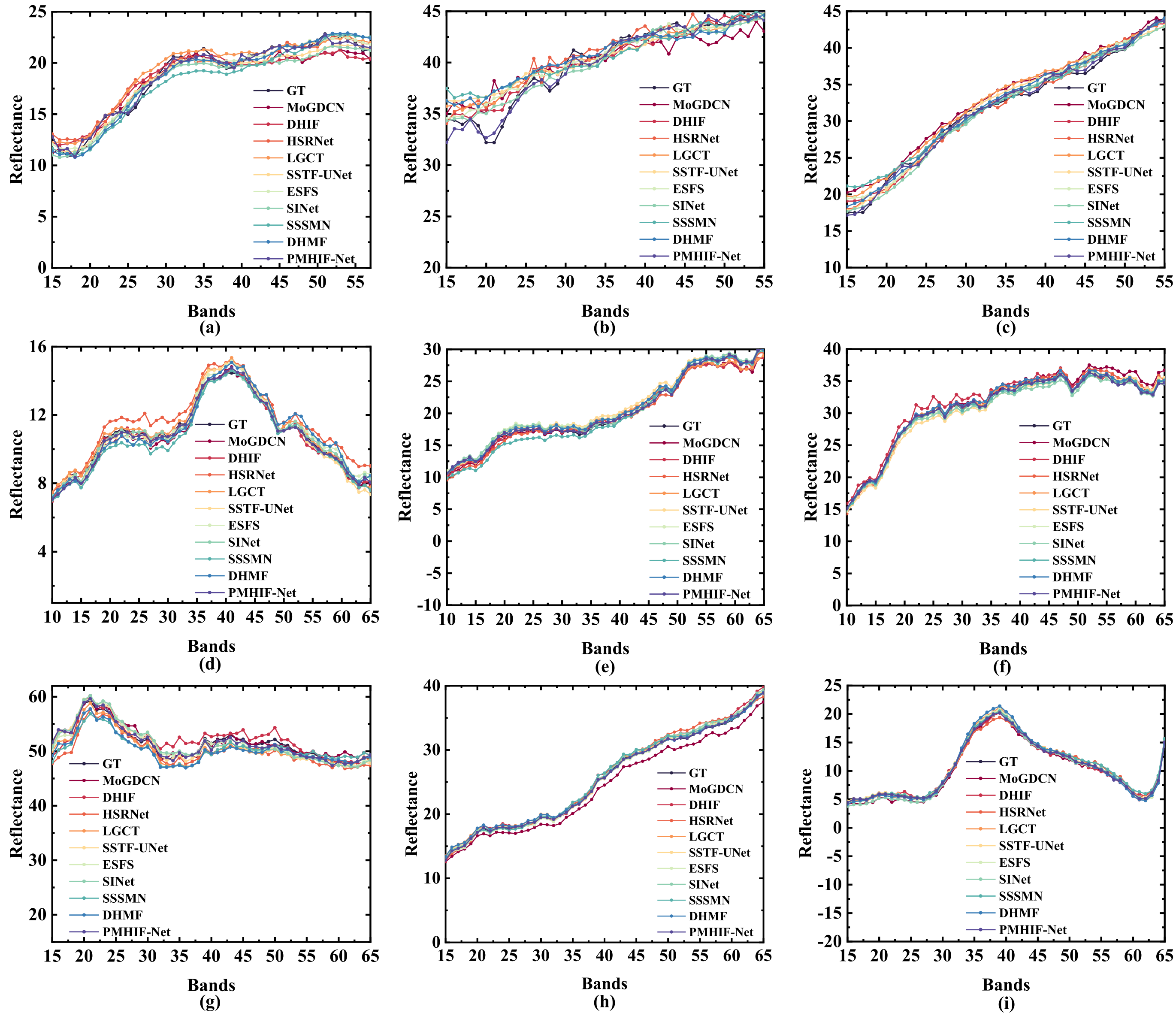

Fig. 10. Spectral curve comparison at random pixel positions on simulated datasets. (a) PaviaU (100, 100), (b) PaviaU (60, 6), (c) PaviaU (16, 96), (d) Houston (64, 64), (e) Houston (66, 93), (f) Houston (48, 90), (g) Chikusei (50, 50), (h) Chikusei (20, 66), and (i) Chikusei (80, 65).

E. Analysis of Experimental Outcomes With Real-World Data

To further verify the effectiveness of our method and evaluate its generalization ability in real and complex scenarios, we conducted comparative experiments with downscaled and full-scale data on Huzhou real dataset. The quantitative analysis results and visualization effects are shown in Table III and Figs. 11 and 12, respectively.

Numerical comparisons verify that the proposed PMHIF-Net attains superior metrics across simulated downscaling tasks and practical full-size real-data tests. In the downscaling experiment, compared to the suboptimal SSTF-UNet, PMHIF-Net improves the PSNR by 0.26 dB, reduces ERGAS by 5.0% and SAM by 3.2%, and the error map (Fig. 11) has the smallest overall deviation, demonstrating a significant advantage in fusion accuracy. Although HSRNet demonstrated competitive fusion performance at the reduced scale, it performed poorly at the full scale. The enlarged red area in Fig. 12 shows that the color in this area significantly deviates from the reference LR-HSI, resulting in spectral distortion. Similar spectral distortion issues are also observed in the DHIF and ESFS methods. However, LGCT and SINet deliver ordinary performance in downscaled experiments but exhibit favorable adaptability in real full-scale scenarios. In the enlarged yellow regions of Fig. 12, MoGDCN and SSSMN suffer from severe spatial blurring and fail to preserve detailed textures consistent with HR-MSI. By contrast, our PMHIF-Net delivers better spatial and spectral fidelity across full real scenes, and its fused outputs closely match the input LR-HSI and HR-MSI. Quantitative results listed in Table III reveal that PMHIF-Net achieves the optimal QNR. Compared with the second-best method, the D_λ and D_s values decrease by 3.1% and 5.5%, respectively. SSSMN obtains a D_λ close to

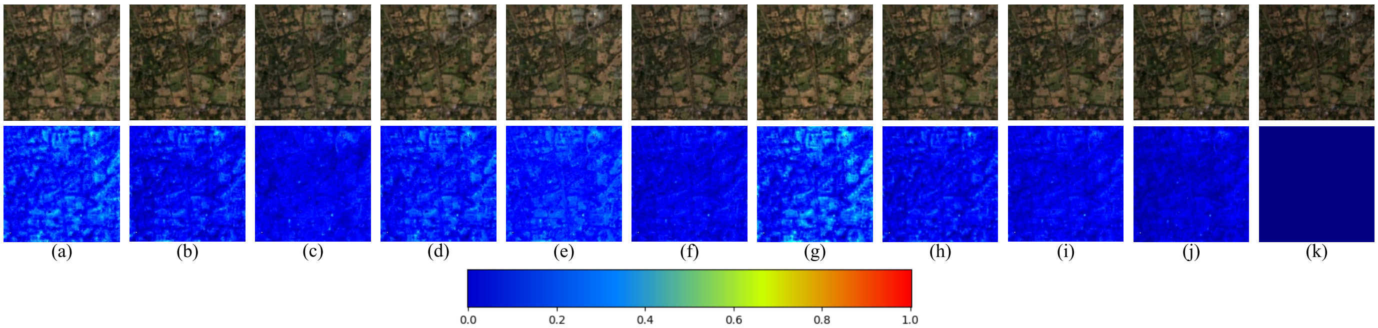

Fig. 11. Visual comparison of reduced-resolution fusion results on Huzhou dataset. The top row presents pseudocolor fused imagery (R-61, G-41, and B-26), and the bottom row displays residual maps against reference images. (a) MoGDCN, (b) DHIF, (c) HSRNet, (d) LGCT, (e) SSTF-UNet, (f) ESFS, (g) SINet, (h) SSSMN, (i) DHMF, (j) PMHIF-Net, and (k) GT.

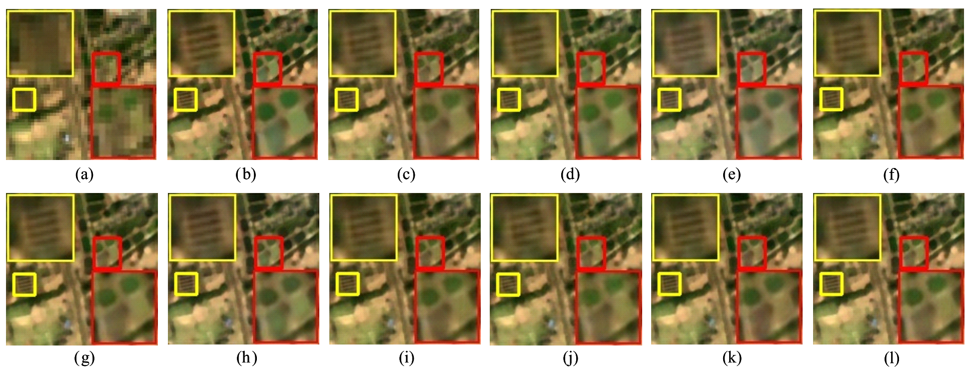

Fig. 12. Visual comparison of full-resolution fusion results on Huzhou dataset (R-61, G-41, and B-26). (a) LR-HSI, (b) HR-MSI, (c) MoGDCN, (d) DHIF, (e) HSRNet, (f) LGCT, (g) SSTF-UNet, (h) ESFS, (i) SINet, (j) SSSMN, (k) DHMF, and (l) PMHIF-Net.

TABLE IV

QUANTITATIVE RESULTS FOR THE EFFECTIVENESS ANALYSIS OF SPALM AND SPELM (BOLD FONTS DENOTE OPTIMAL VALUES)

Methods	PaviaU				Houston				Chikusei			
	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM
Ideal value	$+\infty$	0	0	1	$+\infty$	0	0	1	$+\infty$	0	0	1
SS2D+SS2D	44.5533	1.1112	1.7659	0.9827	51.6612	1.0181	1.1069	0.9927	43.6269	1.5256	1.0915	0.9785
SS2D+SpeLM	44.6306	1.1040	1.7499	0.9828	51.7133	1.0375	1.1025	0.9927	43.6842	1.5281	1.0925	0.9790
SpaLM+SS2D	44.6729	1.0958	1.7447	0.9827	51.7673	1.0148	1.1018	0.9927	43.7203	1.5161	1.0861	0.9792
SpeLM+SpeLM	44.7240	1.0969	1.7283	0.9829	51.8577	1.0075	1.0892	0.9928	43.8221	1.5198	1.0823	0.9795
SpaLM+SpaLM	44.7418	1.0947	1.7248	0.9830	51.9020	1.0019	1.0776	0.9928	43.9050	1.4823	1.0734	0.9801
SpaLM+SpeLM	**44.8173**	**1.0860**	**1.7136**	**0.9831**	**52.1071**	**0.9796**	**1.0777**	**0.9930**	**44.0506**	**1.4041**	**1.0514**	**0.9806**

PMHIF-Net yet yields higher D_s, which implies that SSSMN maintains acceptable spectral fidelity while lacking sufficient perception of spatial details. DHMF-Net achieves comparable D_s to PMHIF-Net but presents higher D_λ. These observations fully demonstrate that PMHIF-Net can simultaneously restrain spatial and spectral distortion under real scenarios. The proposed method exhibits outstanding fusion performance and strong generalization capability for real cross-data applications.

F. Ablation Experiments

Separate ablation tests over three simulated datasets are implemented to validate the validity of every core module and the complete network framework, as well as quantify how each component boosts overall fusion quality.

1) Effectiveness Verification of SpaLM and SpeLM: To verify the practical efficacy of edge prior-guided SpaLM and gradient prior-aware SpeLM in the learning of spatial–spectral features, we adopt SS2D with four-directional scanning as the baseline and conduct ablation experiments by sequentially replacing the core modules of the network. Table IV records all quantitative metrics. Experiments reveal that replacing SS2D in either branch with the proposed SpaLM or SpeLM yields consistent and prominent improvements across all fusion metrics. Checking the results in Fig. 13, the network integrated with SpaLM and SpeLM generates the least bias against reference data and delivers top reconstruction quality. Meanwhile, this article further compares the optimization effect of the edge prior on the extraction of spatial features. Compared with Fig. 13(d), Fig. 13(e) presents more distinguishable

TABLE V
QUANTITATIVE RESULTS FOR THE EFFECTIVENESS ANALYSIS OF SSIMB (BOLD FONTS DENOTE OPTIMAL VALUES)

Methods	PaviaU				Houston				Chikusei			
	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM
Ideal value	$+\infty$	0	0	1	$+\infty$	0	0	1	$+\infty$	0	0	1
w/o CMM	44.6315	1.1028	1.7489	0.9827	51.7546	1.0174	1.0975	0.9927	43.6761	1.4995	1.0831	0.9790
w/o Gate	44.7089	1.1006	1.7303	0.9829	51.8979	1.0096	1.0855	0.9928	43.8659	1.4561	1.0782	0.9797
SSIMB	**44.8173**	**1.0860**	**1.7136**	**0.9831**	**52.1071**	**0.9796**	**1.0777**	**0.9930**	**44.0506**	**1.4041**	**1.0514**	**0.9806**

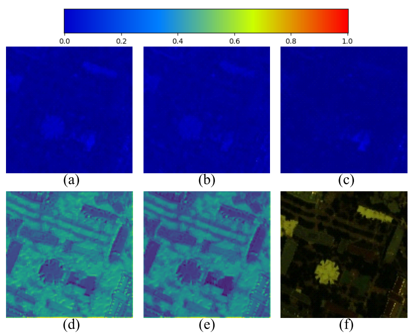

Fig. 13. Visual results of ablation analysis for SpaLM and SpeLM on Houston dataset. (a) SS2D + SpeLM (error map), (b) SpaLM + SS2D (error map), (c) SpaLM + SpeLM (error map), (d) SS2D + SpeLM (feature map), (e) SpaLM + SpeLM (feature map), and (f) GT.

feature responses at the junction of structural edges and flat homogeneous regions, yielding sharper and more complete ground object contours. Such performance improvements can be explained by the unique prior learning mechanism of the designed modules. Specifically, the SpaLM module introduces explicit edge prior information to generate dynamic adaptive edge weights during feature scanning. This mechanism adaptively enhances edge perception along different scanning directions and significantly improves the model's capability of capturing fine-grained spatial details and structural textures. In contrast, the SpeLM module imposes spectral gradient prior constraints on spectral feature learning, which enables the network to precisely identify subtle feature differences and reflectance variations between adjacent spectral bands. Furthermore, the experimental results also indicate that the performance of the combination using dual branches with homogeneous modules (SpaLM + SpaLM and SpeLM + SpeLM) is inferior to that of the network structure of SpaLM + SpeLM in collaboration. Based on the above quantitative and visual results, it can be confirmed that SpaLM and SpeLM can achieve complementary feature learning from two dimensions: spatial edge details and spectral reflectance changes. Working together, they can fully explore the inherent spatial–spectral features and effectively verify the rationality and effectiveness of the design of these two types of modules.

2) Effectiveness Verification of SSIMB: We set up several ablation test groups for metric contrast to confirm SSIMB's ability to boost cross-modal interaction of spatial and spectral features. Table V records all related experimental data. This experiment set up two sets of ablation variants: "w/o CMM" indicates the removal of the cross-modal interaction mechanism, where the original bidirectional mutual-control interactive structure is replaced with a conventional parallel Mamba architecture; "w/o Gate" indicates the removal of the dual-modal dynamic gating modulation branch. As shown in Table V, the removal of either the CMM interaction structure (w/o CMM) or the gating module (w/o Gate) leads to a noticeable degradation in all fusion evaluation metrics. In particular, when the CMM cross-modal interaction mechanism is removed, the model performance decreases most significantly.

These experimental results fully demonstrate the core advantages of the designed SSIMB module. On the one hand, the CMM structure enables bidirectional information interaction and collaborative learning between spatial and spectral features, which effectively breaks the limitation of independent dual-branch feature extraction adopted by most existing methods. On the other hand, the dual-modal dynamic gating modulation adaptively adjusts the relative importance of spatial and spectral feature learning during model inference. It dynamically screens out effective features and suppresses redundant information throughout the fusion process. Benefiting from these two core designs, SSIMB accurately retains critical spatial and spectral details in the stage of in-depth cross-modal fusion, which steadily guarantees the high-precision fusion capability of the overall network.

3) Effectiveness Verification of FSSG: We carry out quantitative ablation tests to verify FSSG's role in feature refinement and redundant signal suppression within PMHIF-Net, with all test metrics listed in Table VI. Three ablation variants are designed for controlled comparison. Specifically, "w/o FSSG" indicates the removal of the FSSG module, "w/o FD" represents the removal of the frequency domain decomposition operation in FSSG, and "w/o SSM" indicates the removal of Mamba in FSSG. According to Table VI, after removing the entire FSSG module (w/o FSSG), the model performance decreased the most significantly. This is because the adaptive feature gating refinement ability is lost, and redundant features cannot be screened and filtered, resulting in chaotic fusion information and decreased accuracy. In comparison with the two incomplete module variants (w/o SSM and w/o FD), the model equipped with the full FSSG structure achieves remarkable performance improvements in all fusion metrics. The visual comparison results in Fig. 14 further confirm this conclusion. The fusion result obtained by fully using the FSSG module has the smallest deviation from the true value image and achieves a better reconstruction effect. Based on the above results, it can be concluded that FSSG constructs an effective feature refinement gating mechanism through

TABLE VI
QUANTITATIVE RESULTS FOR THE EFFECTIVENESS ANALYSIS OF FSSG (BOLD FONTS DENOTE OPTIMAL VALUES)

Methods	PaviaU				Houston				Chikusei			
	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM
Ideal value	$+\infty$	0	0	1	$+\infty$	0	0	1	$+\infty$	0	0	1
w/o FSSG	44.5818	1.1101	1.7599	0.9826	51.7132	1.0375	1.1025	0.9927	43.6397	1.5257	1.0922	0.9789
w/o FD	44.6529	1.0997	1.7502	0.9827	51.7763	1.0238	1.0907	0.9927	43.7524	1.5260	1.0890	0.9792
w/o SSM	44.7348	1.0947	1.7253	0.9830	51.8861	1.0301	1.0894	0.9928	43.8351	1.4739	1.0761	0.9797
FSSG	**44.8173**	**1.0860**	**1.7136**	**0.9831**	**52.1071**	**0.9796**	**1.0777**	**0.9930**	**44.0506**	**1.4041**	**1.0514**	**0.9806**

TABLE VII
QUANTITATIVE RESULTS OF NETWORK DEPTH ANALYSIS ON MODEL PERFORMANCE (BOLD FONTS DENOTE OPTIMAL VALUES)

Stage	PaviaU				Houston				Chikusei			
	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM	PSNR	ERGAS	SAM	SSIM
Ideal value	$+\infty$	0	0	1	$+\infty$	0	0	1	$+\infty$	0	0	1
N=1	44.0495	1.1512	1.8585	0.9818	51.1826	1.0473	1.1879	0.9916	43.1298	1.6450	1.1349	0.9766
N=2	44.5739	1.1054	1.7664	0.9826	51.6925	1.0214	1.1355	0.9926	43.5755	1.6041	1.1121	0.9773
N=3	**44.8173**	**1.0860**	**1.7136**	**0.9831**	**52.1071**	**0.9796**	**1.0777**	**0.9930**	**44.0506**	**1.4041**	**1.0514**	**0.9806**
N=4	44.6657	1.0983	1.7442	0.9828	51.7733	1.0223	1.1137	0.9926	43.6251	1.5967	1.1098	0.9778
N=5	44.4479	1.1158	1.7896	0.9825	51.4589	1.0509	1.1640	0.9923	43.4463	1.6464	1.1322	0.9772

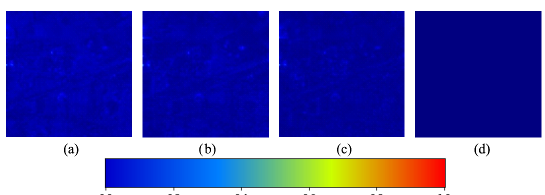

Fig. 14. Visual error maps for FSSG ablation analysis on the Chikusei dataset. (a) w/o FD, (b) w/o SSM, (c) FSSG, and (d) GT.

TABLE VIII
ANALYSIS RESULTS FOR THE EFFECTIVENESS OF SOBEL EDGE PRIORS UNDER NOISY CONDITIONS

Noise	Methods	PaviaU		Houston		Chikusei	
		PSNR↑	SAM↓	PSNR↑	SAM↓	PSNR↑	SAM↓
40dB	EGSSM	44.4306	1.7715	51.4179	1.1801	43.4769	1.1021
	SS2D	44.2761	1.7923	51.1588	1.1971	43.2603	1.1292
30dB	EGSSM	44.2924	1.7884	51.2644	1.1855	42.9573	1.1517
	SS2D	44.1108	1.8026	51.0326	1.2022	42.6358	1.1829
20dB	EGSSM	43.0193	1.9275	49.7635	1.2701	39.5628	1.5583
	SS2D	42.8725	1.9401	49.4050	1.2712	39.1716	1.5852

frequency-domain perception and global sequence modeling. It can precisely retain the effective high- and low-frequency detail features during the image reconstruction stage, while eliminating invalid and redundant information, thereby effectively optimizing the fusion reconstruction effect of the model.

4) Analysis of Network Depth on Model Performance: We carry out contrast tests with varied network depths to analyze how layer configurations affect PMHIF-Net's fusion capability, with all corresponding numerical metrics summarized in Table VII. As can be seen, N represents the number of network layers of PMHIF-Net. The experimental results show that the model performance steadily improves when the layer number N increases from 1 to 3. This is because shallow networks are incapable of sufficient feature extraction and cross-modal interaction. As the number of layers is appropriately increased, the feature learning and fusion modeling capabilities of the network are effectively released. When the number of network layers exceeds 3, the model performance shows a gradual decline trend. This is mainly because excessively deep structures tend to introduce redundant feature information and trigger overfitting risks, which adversely degrade the final image fusion accuracy. Consistent experimental phenomena can be observed on all test datasets, and the network achieves the optimal comprehensive fusion performance when N is set to 3. his finding fully demonstrates the scientificity and rationality of the three-layer network configuration adopted in the proposed PMHIF-Net.

5) Validation of the Effectiveness of Sobel Edge Priors Under Noisy Conditions: Multisource remote sensing images in practical scenarios are inevitably contaminated by noise. Therefore, we verify the effectiveness of guidance based on Sobel edge priors under various noisy environments. Specifically, Gaussian noise with a fixed SNR of 40 dB is added to HR-MSI, while Gaussian noise with SNR values of 40, 30, and 20 dB is injected into LR-HSI to simulate noise environments with different intensities. The experimental results are listed in Table VIII. As shown in Table VIII, model performance gradually declines with the increase of noise intensity, yet favorable fusion performance can still be maintained. Under all noise levels, the scheme equipped with edge prior guidance outperforms the conventional four-directional SS2D. This reveals that the proposed edge prior guidance mechanism possesses certain noise robustness and enables the perception of edge details in noisy environments.

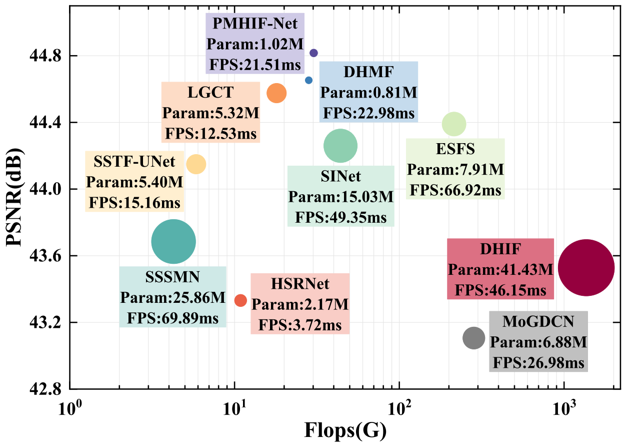

Fig. 15. Bubble chart of model complexity and fusion performance. The horizontal axis denotes FLOPs, the vertical axis represents PSNR, bubble size indicates Param, and colored boxes label parameter amounts and inference time.

G. Model Complexity Analysis

To objectively compare the computational overhead and comprehensive performance of different algorithms, all competing methods are tested under the settings described in Section IV-B on the PaviaU dataset. The input dimensions are LR-HSI of $32 \times 32 \times 103$ and HR-MSI of $128 \times 128 \times 5$. We conduct unified evaluations on parameters (Param), computational complexity (FLOPs), fusion accuracy (PSNR), and inference speed (FPS). The comprehensive comparison results of the above indicators are visualized in Fig. 15. In the figure, the X-axis represents FLOPs and the Y-axis represents PSNR, and the size of each bubble corresponds to the parameter scale of each model, and the exact values of parameters and inference time are labeled within the colored box for each approach. As observed from the statistical results, the proposed PMHIF-Net achieves superior parameter control with relatively low memory overhead. It only increases the parameter number by 0.21 M compared with DHMF, that owns the minimum parameters, and exhibits far fewer parameters than other competing algorithms. In terms of computational complexity, although the FLOPs of PMHIF-Net are slightly higher than those of SSSMN, SSTF-UNet, HSRNet, LGCT, and DHMF, it presents obvious computational efficiency advantages over several complex networks, including SINet, ESFS, MoGDCN, and DHIF. For inference efficiency, PMHIF-Net runs slower than HSRNet, LGCT, and SSTF-UNet, yet achieves faster inference than the second-best method, DHMF, with a reduction of 1.47 ms per inference.

Although PMHIF-Net does not obtain the optimal results in terms of parameters, FLOPs, or inference speed, its fusion accuracy PSNR is the highest among all methods. This fully demonstrates that the elaborately designed network structure of PMHIF-Net achieves significant performance improvement with moderate and controllable computational overhead. PMHIF-Net strikes a favorable tradeoff between calculation speed and reconstruction precision, making the model suitable for real-world engineering deployment.

V. Discussion

As a vital research branch of image fusion, HMIF possesses great practical application value. The quality of fused images directly determines the performance of subsequent downstream tasks, including land cover classification, object detection, and change detection.

For classification tasks, HR-HSI can greatly improve the classification accuracy of various ground objects such as crop types, urban functional zones, and forest species. It provides superior discriminative ability for categories with highly similar spectral characteristics, thereby supporting precision agricultural applications, including crop growth monitoring, early pest and disease identification, and yield prediction. For object detection tasks, HR-HSI integrates fine spatial localization and high-discrimination spectral features, improving the detection accuracy of small-scale targets such as vehicles, ships, and buildings. It can also identify geological features, including mineral alteration zones, lithological boundaries, and soil types, providing reliable basic data for mineral resource investigation. For change detection tasks, high-quality HR-HSI accurately captures spatiotemporal variations in land cover. It provides solid data support for land use monitoring, deforestation tracking, and urban construction analysis, and further enables environmental monitoring tasks such as water quality evaluation, vegetation health assessment, and air pollutant tracking. In addition, it supports disaster impact evaluation, including flood boundary extraction and earthquake damage assessment, as well as urban planning applications such as impervious surface identification and urban heat island analysis.

The proposed PMHIF-Net adopts a prior-guided hierarchical interactive Mamba fusion strategy, which provides an efficient and robust solution for the above downstream remote sensing tasks. Experimental results on the self-built real Huzhou dataset verify that the proposed method effectively improves the spatial resolution of GF-5B images in the study area. In future work, we will apply the proposed model to crop monitoring and wetland monitoring tasks based on local image data to further validate its practical effectiveness.

VI. Conclusion

This article proposes a prior-guided Mamba hierarchical interactive fusion network named PMHIF-Net. The overall architecture consists of four core modules: the SpaLM, the SpeLM, the SSIMB, and the RGRM. In the feature learning stage, the network adopts prior-guided state-space modeling through SpaLM and SpeLM to independently capture spatial edge details and spectral reflectance variation patterns. While maintaining the capability of global long-range dependency modeling, this design effectively strengthens the representation of local spatial features and improves the contextual correlation learning across spectral bands. In the feature interaction, the designed SSIMB is capable of conducting comprehensive cross-modal interaction learning by integrating the extracted spatial and spectral features at each level with the fusion information from preceding layers. Meanwhile, it utilizes the dual-modal dynamic modulation mechanism to achieve adaptive feature fusion, effectively suppressing the interference from redundant features during the fusion process. In the feature reconstruction stage, the network combines multilevel features and utilizes FSSG to achieve Mamba global modeling under frequency-domain prior constraints. It adaptively selects the effective high- and low-frequency details, while retaining the fine features of the ground objects and further eliminating the invalid and redundant components, thereby improving the quality of the reconstructed image. Quantitative and qualitative experiments carried out over PaviaU, Houston, and Chikusei datasets validate the outstanding spatial–spectral reconstruction capability delivered by PMHIF-Net. Moreover, evaluations on the self-built real-scene Huzhou dataset further validate the promising generalization and adaptation ability of the proposed network in practical and cross-data scenarios.

Future research efforts will focus on lightweight and efficient modeling, exploring more optimal selective scanning mechanisms, uncovering more efficient Mamba improvement solutions, and further enhancing the computational efficiency of the model while maintaining the accuracy of HSI fusion. This will enable the model to better adapt to actual remote sensing fusion application scenarios.

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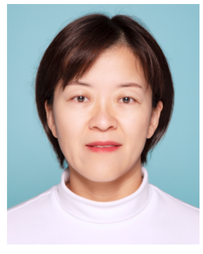

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